

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/04/2025

(21) Application No.202541040855 A

(43) Publication Date : 23/05/2025

(54) Title of the invention : Smart Motion Indicator Helmet

(51) International classification :A42B0003040000, G06F0003160000, B62J0006057000, B60Q0009000000, B23K0009320000

(86) International Application No :NA
Filing Date :NA

(87) International Publication No : NA

(61) Patent of Addition to Application Number :NA
Filing Date :NA

(62) Divisional to Application Number :NA
Filing Date :NA

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(57) Abstract :

The Smart Motion Indicator Helmet (100) is a safety-enhancing wearable designed for motorcyclists and cyclists. It features an outer shell (102) integrated with left (102A) and right (102B) turn indicators, which activate automatically based on the rider's head movements. The system includes motion tracking sensors (201) that detect turns and trigger the corresponding indicator. A head nod recognition system (204) allows for hands-free deactivation of the signal. The helmet is powered by a USB Type-C rechargeable battery (103) for efficient and long-lasting operation. Additionally, a buzzer (104) provides audio feedback, and a cushion (105) ensures rider comfort. The system enhances road safety by improving visibility and eliminating the need for hand signals, reducing distractions while riding. This smart helmet provides a seamless and intuitive way for riders to communicate their movements, making roads safer for everyone.

No. of Pages : 12 No. of Claims : 4

smart motion indicator helmet

Field of Inventions

This invention falls under the smart safety devices and smart wearable devices

Background of Invention

Imagine bustling city streets, where cyclists navigate through traffic, pedestrians, and intersections. Cyclists often face challenges when signaling their intentions to turn or change lanes. Traditional hand signals can be cumbersome, especially in heavy traffic or adverse weather conditions. That's where your invention comes in!

The primary motivation for creating this helmet was to enhance cyclist safety. By integrating turn indicators directly into the helmet, you're providing a more intuitive and efficient way for cyclists to signal their intentions. When a cyclist turns their head to look behind or to the side, the helmet's built-in sensors detect the movement and activate the corresponding turn signal. This minimizes the need for hand signals, allowing cyclists to keep both hands on the handlebars and maintain better control. And also, as an added advantage a beep sound is made whenever the indicators are turned on/off and the rider just need to nod his head forward to turn the indicators off

Beyond cycling, your invention could inspire similar innovations in other areas. Imagine helmets for motorcyclists, scooter riders, or even construction workers—where head movements trigger safety features like brake lights, hazard signals, or communication systems.

Summary of Invention

Bicycle helmet with turn-indicators represents a fusion of safety, technology, and user-friendly design. By addressing real-world challenges faced by cyclists, you're contributing to a safer and more efficient urban mobility experience. Kudos to your inventive spirit!

Abstract

The smart motorcycle indicator helmet enhances safety by using motion sensors to detect head movements. A nod to the right or left activates the corresponding turn indicator, while a nod towards the front turns it off. This hands-free operation allows riders to signal intentions without removing their hands from the handlebars, improving overall road safety.

We Claim

1. Smart Motion Indicator Helmet consists of various components, An alarm system with a buzzer (104) and 2 led indicators left(101) and right(102), it also have a type-c USB port(103) which is used to recharge the battery to turn on the indicators;
2. The system claim 1, Audible Feedback (104) which allows the rider to know the status of the indicator whether it is on or off. It helps to avoid the hustle of looking backward to know the indicator status;
3. The system claim 1, The battery can be recharged with USB-C cable in its respective USB port(103) and also providing long battery life up to 3 days of continuous use;
4. The system claim 1, The outer shell of the helmet is made of composite fiberglass, polycarbonate material for sudden impacts and ensure structural integrity;